

Electron-coupling-to-waveguide

Transverse energy estimation

$$E_{\perp} = E_{\text{electron}} \theta^2$$

$$\Delta E_{\perp} = 2E_{\text{electron}} \theta \Delta \theta$$

For smaller d , the slop is bigger

$$W = 400 \text{ nm}$$

$$h = 300 \text{ }\mu\text{m}$$

$$E_e = 200 \text{ keV}$$

